School of Science, Computing and Emerging Technologies

COS30043 Interface Design and Development

Project: Building a Modern Web Application

1. Overview

This project is designed to help you develop practical skills in building modern web applications using Vue.js 3 and the Bootstrap framework. Through completing this project, you will develop and apply your skills in structuring an application, implementing interactive features, and following best practices in web development.

In this project, you have the freedom to choose the theme of your application, such as an e-commerce site, event management system, educational course platform, or real estate listing website. You are not limited to these examples; you may choose any theme of interest. Regardless of the theme you choose, your application must meet the following requirements.

2. Requirements

2.1 Functional Requirements

- The system must consist of at least 10 interconnected pages with meaningful navigation between them.
- The system must display a collection of items (such as products, courses, articles, or events) and allow users to view detailed information for each item.
- Users must be able to search for content using various criteria and sort the results in different ways.
- The system must support user registration and login, with different content visibility for authenticated and unauthenticated users.
- Authorised users must be able to create, edit, and delete content.
- The system must include social interaction features, such as liking or voting on content.
- The system must store user and application data persistently so that information remains available across sessions.

2.2 Technical Requirements

- The project must use Vue.js (version 3) and the Bootstrap grid system. It should be created using Vite and deployed on Mercury.
- The application must follow a mobile-first approach and be fully responsive across at least three device sizes.
- The system must utilise a component-based architecture in Vue.js to ensure a modular and maintainable design.
- The system must use Vue Router to manage navigation between pages.

- The system must demonstrate effective use of core Vue.js features, including directives (e.g., v-bind, v-model, v-if, v-for, v-on), methods, and computed properties to create interactive and reactive user interfaces.
- The system must implement pagination for data display.
- The system must implement forms with appropriate data validation.
- The system must include a backend database system to support the frontend, including a RESTful API to handle CRUD operations.
- The system must apply coding conventions consistently, including proper naming, and indentation for readability and professionalism.

2.3 Advanced Features

Every student is encouraged to independently learn and implement one advanced feature using techniques not covered in class. The feature must be related to Vue.js and should significantly enhance the application. The feature must be clearly demonstrated in the video and explained in the individual report.

3. Project Report (group report, no more than 12 pages)

The report should include at least the following sections:

- The main functionality of your application and the techniques used.
- Relevant screenshots of the application.
- Highlights of any innovative features or unique approaches implemented (if applicable).
- An evaluation of the usability and accessibility of the interface, along with a discussion of potential improvements.

4. Demonstration Video

Each group is required to submit a 5–10 minute video demonstrating all functionalities of the application. Each member should contribute a similar amount of presentation time. When a member is speaking, their face must be clearly visible on camera. The demonstration must be recorded in a single continuous take (without cuts, edits, or post-production).

Please host your video on a suitable platform (e.g., YouTube, OneDrive) and submit only the video link via Canvas. Ensure that the video is accessible to others.

5. Individual Report (no more than 5 pages)

Each student is required to submit an individual report outlining their contributions, the key techniques used, the challenges encountered, and how they were resolved. If applicable, the report should also include a description of the advanced feature implemented, including the rationale for its inclusion and the techniques used.

6. Submission

Please submit the following via Canvas by the specified due date:

- Source code (to be submitted by one group member)
- Mercury URL to your project (to be submitted by the same member)
- Group report (to be submitted by the same member)
- Demonstration video link (to be submitted by the same member)
- Individual report (to be submitted by each member)
- Peer evaluation form (to be submitted by each member)

7. Marking Scheme

Marks are based on the implementation of key components and students' understanding of the required knowledge and techniques.

Assessment Criteria	Mark
The Application (based on the onsite demonstration, demonstration video, and source code)	
1) Functional Requirements • All functional requirements are fully met. The application demonstrates sufficient complexity for real-world use. (4-5 marks) • Basic functionality is implemented but requires refinement to meet real-world usage standards. (2-3 marks) • Basic functionality is missing or poorly implemented. The application is difficult to use. (0–1 mark)	5
2) Technical Requirements • All technical specifications are fully implemented. (4–5 marks) • Most technical specifications are met, but minor errors or issues are present. (2–3 marks) • Major technical issues or missing elements are present. (0–1 mark)	5
3) Advanced Features and Innovation • The technique is advanced, relevant, and significantly enhances the application. (7–10 marks) • The technique is moderately advanced and somewhat relevant. (4–6 marks) • The technique is basic or poorly integrated into the application, with little evidence of self-learning. (0–3 marks)	10
Project Report • Clear, well-structured report that covers all required elements. (4–5 marks) • Report covers basic elements but lacks depth. (2–3 marks) • Report is poorly written or lacks critical information. (0–1 mark)	5
Individual Report • Clear, well-structured report that covers all required elements. Contribution is	5

substantial. (4–5 marks) • Report covers basic elements but lacks depth. Contribution is moderate or limited. (2–3 marks) • Report is poorly written or lacks critical information. Contribution is minimal or unclear. (0–1 mark)	
Interview During the Week 12 tutorial, the tutor will interview you about your project and the techniques used in its implementation.	10
Total	**40**

Your group mark (app and project report) will be adjusted based on peer evaluation.
Your final total mark will also be adjusted according to your interview performance.